Experiment No. 3

[i] bubble sort :

```
#include <stdio.h>

int main() {
    int arr[100], n, i, j, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for(i = 0; i < n-1; i++) {
        for(j = 0; j < n-i-1; j++) {
            if(arr[j] > arr[j+1]) {
                temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }

    printf("Sorted array:\n");
    for(i = 0; i < n; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

```
C:\Users\PRASANNA\Desktop\DS practicals\4th practical>cd "c:
th practical\"bubble
Enter number of elements: 10
Enter elements:
12
32
11
45
65
98
123
234
4
54
Sorted array:
4 11 12 32 45 54 65 98 123 234
c:\Users\PRASANNA\Desktop\DS practicals\4th practical>
```

[ii] insertion :

```
#include <stdio.h>

int main() {
    int arr[100], n, i, j, key;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for(i = 1; i < n; i++) {
        key = arr[i];
        j = i - 1;

        while(j >= 0 && arr[j] > key) {
            arr[j+1] = arr[j];
            j--;
        }

        arr[j+1] = key;
    }

    printf("Sorted array:\n");
    for(i = 0; i < n; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

```
c:\Users\PRASANNA\Desktop\DS practicals\4th practical>cd
cals\4th practical\"insertion
Enter number of elements: 10
Enter elements:
34
12
24
56
3
7
12
67
98
44
Sorted array:
3 7 12 12 24 34 44 56 67 98
c:\Users\PRASANNA\Desktop\DS practicals\4th practical>
```

[iii] Merge :

```
#include <stdio.h>

void merge(int arr[], int l, int m, int r) {
    int i, j, k;
    int n1 = m - l + 1;
    int n2 = r - m;

    int L[50], R[50];

    for(i = 0; i < n1; i++)
        L[i] = arr[l + i];
    for(j = 0; j < n2; j++)
        R[j] = arr[m + 1 + j];

    i = 0; j = 0; k = l;

    while(i < n1 && j < n2) {
        if(L[i] <= R[j]) {
            arr[k] = L[i];
            i++;
        } else {
            arr[k] = R[j];
            j++;
        }
        k++;
    }

    while(i < n1) {
        arr[k] = L[i];
        i++; k++;
    }

    while(j < n2) {
        arr[k] = R[j];
        j++; k++;
    }
}

void mergesort(int arr[], int l, int r) {
    if(l < r) {
        int m = (l + r) / 2;

        mergesort(arr, l, m);
        mergesort(arr, m+1, r);
        merge(arr, l, m, r);
    }
}
```

```
c:\Users\PRASANNA\Desktop\DS pract
th practical\"mearge
Enter number of elements: 10
Enter elements:
43
56
11
787
343
23
45
76
10
1
Sorted array:
1 10 11 23 43 45 56 76 343 787
c:\Users\PRASANNA\Desktop\DS pract
```

```
int main() {
   int arr[100], n, i;

   printf("Enter number of elements: ");
   scanf("%d",&n);

   printf("Enter elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&arr[i]);

   mergesort(arr,0,n-1);

   printf("Sorted array:\n");
   for(i=0;i<n;i++)
      printf("%d ",arr[i]);

   return 0;
}
```

[iv] quick :

```
#include <stdio.h>

void quicksort(int arr[], int low, int high) {
   int i = low, j = high, pivot, temp;

   pivot = arr[(low + high) / 2];

   while(i <= j) {
      while(arr[i] < pivot)
         i++;

      while(arr[j] > pivot)
         j--;

      if(i <= j) {
         temp = arr[i];
         arr[i] = arr[j];
         arr[j] = temp;
         i++;
         j--;
      }
   }

   if(low < j)
      quicksort(arr, low, j);
   if(i < high)
      quicksort(arr, i, high);
}

int main() {
   int arr[100], n, i;

   printf("Enter number of elements:
");
   scanf("%d",&n);

   printf("Enter elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&arr[i]);

   quicksort(arr,0,n-1);

   printf("Sorted array:\n");
   for(i=0;i<n;i++)
      printf("%d ",arr[i]);

   return 0;
}
```

```
c:\Users\PRASANNA\Desktop\DS practicals\4th practical>cd
 practical\"quick
Enter number of elements: 10
Enter elements:
1
432
23
21
67
45
90
345
12
56
Sorted array:
1 12 21 23 45 56 67 90 345 432
c:\Users\PRASANNA\Desktop\DS practicals\4th practical>
```

[v] selection :

```
//Prasanna Pradeep Bhurke
//Roll No. : 155

#include <stdio.h>

int main() {
    int arr[100], n, i, j, min, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for(i = 0; i < n-1; i++) {
        min = i;
        for(j = i+1; j < n; j++) {
            if(arr[j] < arr[min])
                min = j;
        }

        temp = arr[i];
        arr[i] = arr[min];
        arr[min] = temp;
    }

    printf("Sorted array:\n");
    for(i = 0; i < n; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

```
c:\Users\PRASANNA\Desktop\DS practicals\4th p
cals\4th practical\"selection
Enter number of elements: 10
Enter elements:
12
32
34
76
11
232
1
4546
13
56
Sorted array:
1 11 12 13 32 34 56 76 232 4546
c:\Users\PRASANNA\Desktop\DS practicals\4th p
```